MIL-L-15019E
8 March 1982
SUPERSEDING
MIL-L-0015019D(SH)
13 March 1981
MIL-L-15019C
20 February 1962
(See 6.5)

MILITARY SPECIFICATION

LUBRICATING OIL, COMPOUNDED

* This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lubricating oils compounded with various fatty oils in various percentages.

* 1.2 Classification. The oil shall be of the following Military symbols, as specified (see 6.2):

Military symbol
4065
6135

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Standards. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-791 - Lubricants, Liquid Fuels, and Related Products; Method of Testing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9150

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MILITARY

MIL-STD-290 - Packaging of Petroleum and Related Products.

(Copies of standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 91 - Precipitation Number of Lubricating Oils, Test Method for.
D 92 - Flash and Fire Points by Cleveland Open Cup, Test Method for. (DoD adopted)
D 94 - Saponification Number of Petroleum Products, Test Method for. (DoD adopted)
D 95 - Water in Petroleum Products and Bituminous Materials by Distillation, Test Method for. (DoD adopted)
D 97 - Pour Point of Petroleum Oils, Test Method for. (DoD adopted)
D 129 - Sulfur in Petroleum Products (General Bomb Method), Test Method for. (DoD adopted)
D 130 - Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test, Method for. (DoD adopted)
D 270 - Sampling Petroleum and Petroleum Products, Method for. (DoD adopted)
D 287 - API Gravity of Crude Petroleum and Petroleum Products (Hydrometer Method), Test Method for. (DoD adopted)
D 445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity), Test Method for. (DoD adopted)
D 482 - Ash from Petroleum Products, Test Method for. (DoD adopted)
D 524 - Ramsbottom Carbon Residue of Petroleum Products, Test Method for. (DoD adopted)
D 974 - Neutralization Number by Color-Indicator Titration, Test Method for. (DoD adopted)
D 1401 - Emulsion Characteristics of Petroleum Oils and Synthetic Fluids, Test Method for.
D 1500 - ASTM Color of Petroleum Products (ASTM Color Scale), Test Method for. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample of military symbol 4065 shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Material.

3.2.1 Military symbol 4065. The oil shall be a refined petroleum oil compounded with either blown rapeseed oil, a substitute fatty oil, or with an additive type emulsifying agent in the percentage specified in table I.

3.2.2 Military symbol 6135. The oil shall be a refined petroleum oil compounded with an acidless tallow oil, and in the percentage specifie in table I.

3.3 Chemical and physical requirements. Compounded lubricating oil shall conform to the requirements as specified in table I.

* TABLE I. Chemical and physical requirements.

Characteristics	Symbol 4065	Symbol 6135
Specific gravity 60/60 1/	Report	Report
A.P.I. degrees	Report	Report
Viscosity, centistokes at 100°C	11.2 - 15.6	------
Viscosity, centistokes at 40°C	------	414 - 506
Viscosity index minimum (min)	------	90
Pour point, °C (°F) maximum (max)	-12 (10)	-6 (20)
Flash point, °C (°F) (min)	177 (350)	246 (475)
Fatty oils, percent	10 - 20 2/	5 - 7
Nature of fatty oils	Report 2/	3/
Emulsion with distilled water at 54°C (130°F)	4/	------
Emulsion with 1 percent salt solution at 54°C (130°F)	4/	------
Color	Report	Report
Neutrality, qualitative	Neutral	Neutral
Neutralization number (max)	3.00	0.75
Corrosion, copper strip at 100°C 5/	No. 3 (max)	No. 3 (max)
Water, percent (max) 6/	None	None
Ash, percent (max)	Report	0.05
Carbon residue, percent (max)	1.00	2.50
Total sulfur, percent (max)	1.00	1.00
Precipitation number (max)	.05	.05

See footnotes at top of next page.

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1/ May be converted from A.P.I. degrees.
2/ Blown rapeseed oil or a substitute emulsifier where adequate performance data are available, is acceptable. The 10 - 20 percent fatty oil requirement applies only to blown rapeseed oil or substitute fatty oil. No minimum percentage requirement is established for additive type emulsifying agents, but the finished oil in which such substitutes are used must be of equal performance to and compatible with an oil blended with 10 - 15 percent blown rapeseed oil.
3/ Acidless tallow oil or equivalent.
4/ Shall remain emulsified at end of the 60 minutes.
5/ There shall be no evidence of corrosion or pitting.
6/ No moisture visible in distillation traps.

3.4 Military symbol 4065.

3.4.1 Compatibility. An oil blended with blown rapeseed oil, a substitute fatty oil, or an additive type emulsifying agent shall be compatible with reference oil(s) designated by the Naval Sea Systems Command (NAVSEA), when the additive type to be supplied is specified by the contractor. An oil containing a substitute fatty oil or an additive type emulsifying agent will be considered compatible if, when tested in accordance with 4.3.2.1, values obtained are not less than those of the poorer of the two oils contained in the mixture.

3.5 Homogeneity. Additive agents, shall remain uniformally distributed throughout at all temperatures above the pour point and up to 121°C (250°F). When the oil is cooled below the pour point it shall regain homogeneity on arriving at temperatures of 6°C (10°F) above the pour point.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 First article inspection. First article inspection shall apply only to military symbol 4065 and shall consist of the examination and tests specified in 4.3.1 and 4.3.2.

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* 4.3.1 First article test procedures. First article tests shall be conducted in accordance with ASTM methods as follows:

Test	ASTM Method No.
Specific gravity	D 287
A.P.I. degrees	D 287
Viscosity, kinematic	D 445
Pour point	D 97
Flash point	D 92
Fatty oils	D 94
Emulsion with distilled water	D 1401
Color	D 1500
Neutrality, qualitative	1/
Neutralization number	D 974
Corrosion, copper strip at 100°C	D 130
Water	D 95
Ash	D 482
Carbon residue	D 524
Total sulfur	D 129
Precipitation number	D 91
Emulsion with 1 percent salt water	D 1401
Petroleum products, sampling	D 270

1/ FED-STD-791.

4.3.2 Compatibility.

4.3.2.1 Oil containing blown rapeseed oil. A mixture in the proportion of one part of the oil being tested and one part of reference oil(s) designated by NAVSEA shall be subjected to the following tests:

Test	ASTM Method No.
Emulsion with distilled water	D 1401
Emulsion with 1 percent salt water	D 1401
Precipitation number	D 91

4.4 Sampling for quality conformance.

4.4.1 Inspection lot. For purpose of sampling, a lot shall consist of all compounded lubricating oil of the same Military symbol manufactur as one batch.

4.4.2 Sampling procedures. Test samples shall be obtained from each lot of finished oil in accordance with ASTM D 270.

4.4.2.1 Sampling while filling containers. A sample shall be drawn at the discharge pipe where it enters the receiving vessel or vessels. The total sample shall be not less than 1 quart and shall be composed of small samples of not more than 1 pint each, taken at regular intervals during the entire period of loading or filling.

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4.4.2.2 Sampling from filled containers. From each inspection lot, two containers shall be selected at random. From each of the two containers, two one-quart specimens shall be taken for the tests specified in 4.5.

4.5 Quality conformance inspection.

4.5.1 Inspection shall be in accordance with method 9601 of FED-STD-791.

4.5.2 Examination of filled containers. Filled containers shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Containers shall also be weighed to determine the amount of the contents. Any container having one or more defects or under required fill, shall be rejected.

4.5.2.1 Military symbols 4065 and 6135. The samples selected as specified in 4.4.2 shall be tested as specified in 4.3.1.

4.5.2.2 Rejection of lots. If the samples of lubricating oil fail in any of the tests, the entire lot shall be rejected.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 Packaging and packing. Compounded lubricating oil shall be packaged and packed for the level specified (see 6.2), in accordance with MIL-STD-290. The standard size of the containers are 5 gallon cans and 55 gallon drums, and shall be specified by the contracting activity (see 6.2).

6. NOTES

6.1 Intended use. Military symbol 4065 is used primarily as a stern tube lubricant. Military symbol 6135 lubricating oil is used in special applications, generally involving moisture or worm gears.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Classification of military symbol required (see 1.2).
(c) When first article sample is required (see 3.1).
(d) Size and type of container required (see 5.1).
(e) Selection of the applicable level of packaging and packing required (see 5.1).
(f) Quantity in gallons. The material should be purchased by volume, the unit being a U.S. gallon at 15.6°C (60°F) (see 5.1).

6.3 First article inspection. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which

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has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Superseding information. For U.S. Navy applications, symbol 7105 is superseded by symbol 6135. Symbol 8190 is superseded by MIL-L-17331, symbol 2190 TEP except for worm gear applications where symbo 6135 shall be used.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liabilit whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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